

P4: token-share mix after fresh TL parent (nanochat, TL-39/WikiText-103) (#307591)

Author(s)

This pre-registration is currently anonymous to enable blind peer-review.
It has one author.

Pre-registered on:

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1) Have any data been collected for this study already?

It's complicated. We have already collected some data but explain in Question 8 why readers may consider this a valid pre-registration nevertheless.

2) What's the main question being asked or hypothesis being tested in this study?

This study was designed after released P3 findings and after the P3 B3 document/byte-share ambiguity were known (P3 Gate X 2026-08-20; #307342). It is a post-P3 token-share mixture trade-off preregistration. It does not amend #306780, #306935, or #307342. It is not "P3 B3 fixed."

After a newly trained P4 Tagalog nanochat parent (fresh weights; frozen P1.1 WikiText-TL-39 train; carry-forward P3 Tagalog 32768 BPE) passes P0-T, freeze d20 as C0. Continue C0 for one shared token budget on C1 extra Tagalog, C2 pure English, and C3 a pre-frozen EN/TL mix with $q_{TL}=0.50$ Tagalog source-content tokens under the carry-forward P3 tokenizer (no BOS/pad/pack/crop). Does C3 retain more Tagalog than C2 while still acquiring more English than C1?

$R_{TL} = TL \text{ val_bpb_full}(C3) - TL \text{ val_bpb_full}(C2)$. Success: $R_{TL} \leq -0.01$. $A_{EN} = EN \text{ val_bpb_full}(C3) - EN \text{ val_bpb_full}(C1)$. Success: $A_{EN} \leq -0.01$. Equality at -0.01 counts. Both required for the joint trade-off. No composite. P4 does not claim byte balancing.

P0-T, before any child token, at BOTH d8 and d20: Tagalog val_bpb_full is ≥ 0.01 below untrained same-depth and below a train-fitted add-1 UTF-8 byte unigram. If either depth fails either floor, do not train C1/C2/C3. d8 is eligibility only; d20 is the only child parent.

Do not load P1.1, P2, or P3 weights. Not CORE/chat/SFT. One seed. Mitigation is not claimed here.

3) Describe the key dependent variable(s) specifying how they will be measured.

Primary DVs: Tagalog val_bpb_full on frozen P1.1 val SHA256 4d51644b84d05050bfc8c515079e60f6e437082b6cce2122e9ed00e7b1db2b1c; English val_bpb_full on WT103-raw val SHA256 874dec29844b3d46fc39e5479ee2dc4b3ba37309d9baf3bba4b5654697f3ae3b. Same carry-forward P3 Tagalog BPE on every arm. BPB = mean token NLL / ($\ln(2)$ x mean UTF-8 bytes per token), official evaluate_bpb, T=2048, BOS-bestfit, full val, CUDA. meta.val_bpb and C3 val.parquet are not DVs.

Six child cells: C1/C2/C3 x {TL, EN}. R_{TL} and A_{EN} as in Q2. At Gate U, collect C0 English val once, descriptive, excluded from R_{TL} and A_{EN} . P0-T covers C0 Tagalog (CUDA-only status).

After val seal (test_access=0), one C3-only event on EN test SHA256 2bccabc020cbb8d09273ccdc42ed926957b83824ca767c96fb588041b8d434e and TL test SHA256 3bd193458f4c494d84dae345548c0c01cb6cd7275e98d6ed39a41d517a093baf. C1/C2 never tested. Not a test-set R_{TL}/A_{EN} . Do not reuse 1.164768 or P2/P3 Gate V numbers.

4) How many and which conditions will participants be assigned to?

No human participants. Pin 92d63d4; base_train -> evaluate_bpb; T=2048; B=65536; CORE off. NVIDIA CUDA (A40 class). Never load P1.1/P2/P3 .pt.

Parent: fresh init on P1.1 Tagalog train SHA256 2b0474c5700dc1eba14def572aa23cc227e4c59c10c2de3ce6b7bda75d137687; carry-forward P3 tokenizer.pkl SHA256 04436b854e0841025a3dd2b46baae07a7ccc252e9f99a19171306f00bc5a8 and token_bytes.pt SHA256 a5dbc1c88f6292696108263072d77115718cc2d8357f7ad4859adfa517cc2132; d8+d20; $N_{TL0}=294$. C0 = freeze d20.

From C0, three siblings, $D_{\text{phase2}} = 294 \times 65536 = 19267584$, fresh Muon+AdamW (load_optimizer=False), peak LR = 0.3 x parent peak LR, warmup 14, terminal ckpt only: C1 extra Tagalog. C2 WT103-raw train. C3 $q_{TL}=0.50$ Tagalog source-content tokens (9633792/9633792), frozen after tokenizer freeze and before any P4 BPB; seed 42; 2048-token blocks. Serial C1 then C2 then C3. One seed.

5) Specify exactly which analyses you will conduct to examine the main question/hypothesis.

P0-T at d8 and d20 before any child token (CUDA-only status). If either floor fails, do not train C1/C2/C3.

Primary: six-cell val seal then R_{TL} and A_{EN} as in Q2. Conclusions: both / only-R / only-A / neither. No composite or best-arm. C0 English val descriptive only.

One seed: point estimates; no CI or p. $|\delta| < 0.01$ is not a ranking. Do not retune $q_{TL}/D/\delta/\text{test}$ from P3. After U: Gate-V policy as specified in Q3. Exploratory only: byte-balanced mix, extra seeds, replay, FilBench, EWC, CORE, SFT.

6) Describe exactly how outliers will be defined and handled, and your precise rule(s) for excluding observations.

Official JSONLs are byte-identical frozen copies (train SHA256 2b0474c5... and 09ae691c...; val/test as in Q3). Hash mismatch = stop. No reclean. No split substitution. Do not drop documents to chase byte share.

Exclude a run for: NaN/Inf; ClimbMix; val/test in train; inexact C3 packed quota; wrap before N x B; new tokenizer after BPB; ratio -1; HF Trainer; writing into p1/p2/p3 Hub; P1.1/P2/P3 weights as parent; child used as parent; MPS/CPU for official GPU gates; missing terminal ckpt. Partial child: quarantine; clean restart from C0. Do not drop an arm because a contrast looks bad. d4 smoke is not confirmatory.

7) How many observations will be collected or what will determine sample size? No need to justify decision, but be precise about exactly how the number will be determined.

Frozen P1.1 TL-39 train and official WT103-raw train, no added docs. Parent: one seed, d8 and d20, N=294. Children: one seed each at d20, N=294, D_phase2=19267584. Six confirmatory val cells. After val seal: one C3-only test event (two reads). C1 and C2 are not tested. No extra seeds or ratios.

8) Anything else you would like to pre-register? (e.g., secondary analyses, variables collected for exploratory purposes, unusual analyses planned?)

Q1 is "It's complicated": public corpora and P1.1/P2/P3 outcomes exist; no P4 parent, children, val_bpb_full, or test_bpb has been computed. This PDF locks those unobserved P4 quantities. New study; does not amend #306780, #306935, or #307342. C3 is not P3 B3. P3 magnitudes are not P4 calibration targets. Lockbox release after val seal plus C3-only test, except blocked-P0-T. Exploratory only: byte-balanced mix, extra seeds, replay, FilBench, EWC, CORE, SFT. Master SHA256 22c28f2bc632f132d9c95bbbcc9d1facbddf0b6b821445487e451c472ea58d4b. Gate-bible SHA256 b389b70e0b8e3af869e8dea314b1c7c6b91df313e49d1bf11d9d07961b4a5a42. Pre-filing addendum SHA256 f056a6f75c73a4d8dc3401ba8d7219d406aa7e498e5b0799d3d0373f9f74c216. Pin 92d63d4e8bb4df75c3b71618f31ddde2378b2bcd. The SHA-bound addendum fixes seeds, exact evaluator/packing, terminal-checkpoint rule, tokenizer artifacts, packed-stream quota audit, and CUDA-only P0-T status. Test files stay outside train dirs.

BUNDLE

This pre-registration is part of a bundle which includes:

#306,780 - <https://aspredicted.org/6r6v4v.pdf> - Title: 'NANOCHAT-FILIPINO P1.1 - WikiText-TL-39 fixed-budget depth vs held-out BPB' #307,342 - <https://aspredicted.org/wd2pc8.pdf> - Title: 'P3: TL retention after EN continuation (nanochat, TL-39 then WikiText-103)' #306,935 - <https://aspredicted.org/xa56bs.pdf> - Title: 'P2: EN retention after TL continuation (nanochat, WikiText-103 then TL-39)'